WebClient Computing

HTML Basics

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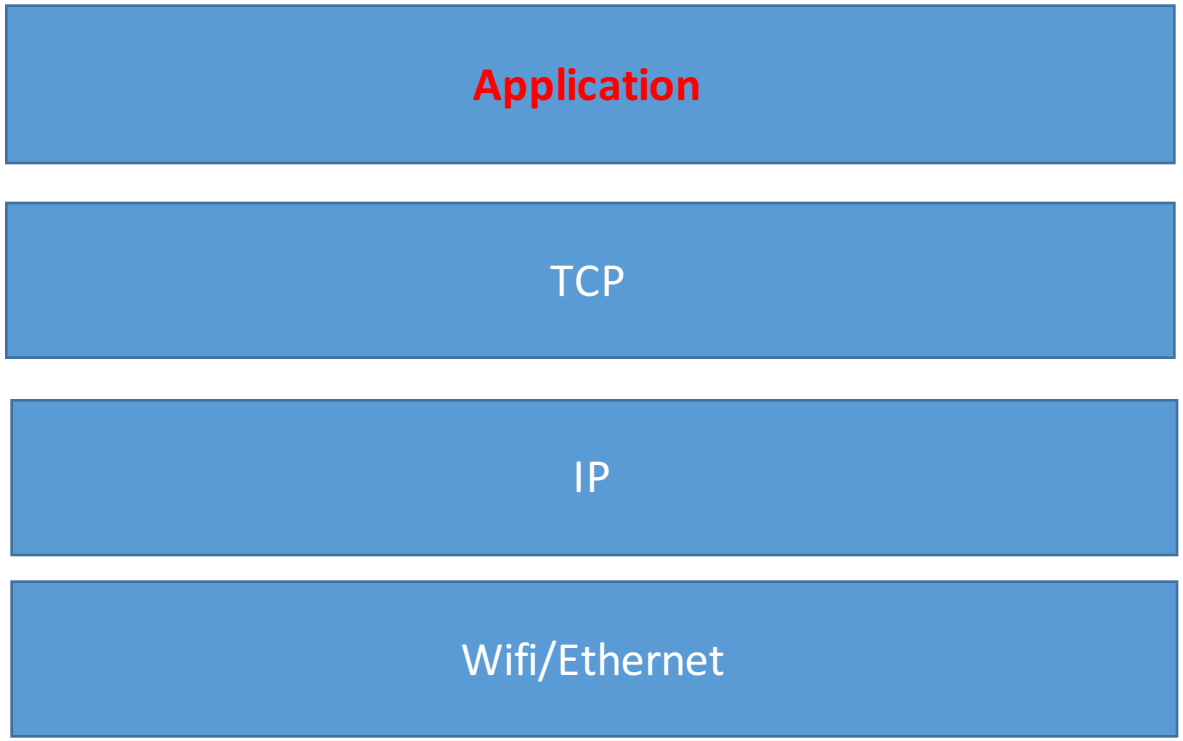
Notice and Follow-up from the Previous Class

- Textbook
  - No textbook (yet – if you all want, I can recommend you one)
  - IMO, Programming is not about learning/reading books/memorizing
  - Programming is about DOING
- Server-client and P2P model

Communication Model

- Server-client model
  - Server: Content provider
    - Naver, Daum, Google, Facebook
  - Client: Content consumer
    - Internet explorer, cell-phone browser
  - Classical approach and straightforward
  - Shortcomings
    - Scalability
    - Single-point-of-failure
- P2P model
  - Each device is a server and client
  - BitTorrent
  - Quality of Service uncertainty

Overview of What We Discussed



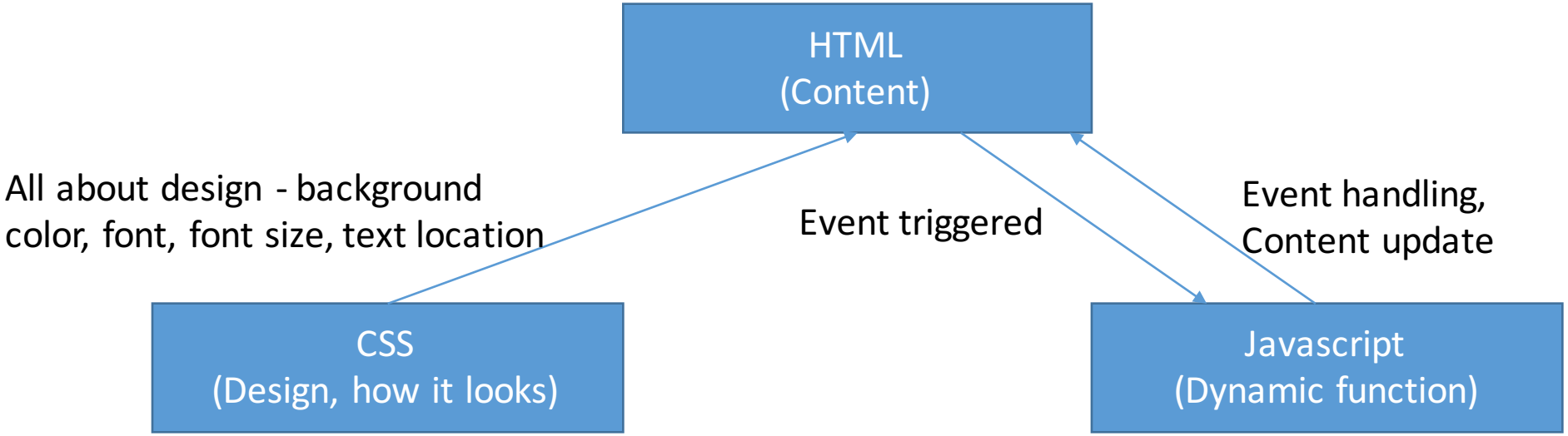
HTML

- **H**yper **T**ext **M**arkup **L**anguage
  - Hyper text
    - a structured text that uses links to navigate through different pages
  - Markup language
    - Represents structure of documents or data
    - Different from programming language that executes with a computer (Java)
- Express title, sentence, list, link, references, …

Example of HTML

```html
<!DOCTYPE html>
<html>
<head>
    <title>My first web page</title>
    <!-- This is a comment, by the way -->
</head>
<body>
<h1>My first web page</h1>
<h2>What this is</h2>
<p>A simple page put together using HTML. <em>I said a simple page put together using HTML.</em> A simple page put together using
page put together using HTML. A simple page put together using HTML. A simple page put together using HTML. A simple page put tog
HTML.</p>
<h2>Why this is</h2>
<ul>
    <li>To learn HTML</li>
    <li>
        To show off
        <ol>
            <li>To my boss</li>
            <li>To my friends</li>
            <li>To my cat</li>
            <li>To the little talking duck in my brain</li>
        </ol>
    </li>
    <li>Because I have fallen in love with my computer and want to give her some HTML loving.</li>
</ul>
<h2>Where to find the tutorial</h2>
<p><a href="http://www.htmldog.com"><img src="http://www.htmldog.com/badge1.gif" width="120" height="90" alt="HTML Dog"></a></p>
<h3>Some random table</h3>
<table>
    <tr>
        <td>Row 1, cell 1</td>
        <td>Row 1, cell 2</td>
        <td>Row 1, cell 3</td>
    </tr>
```

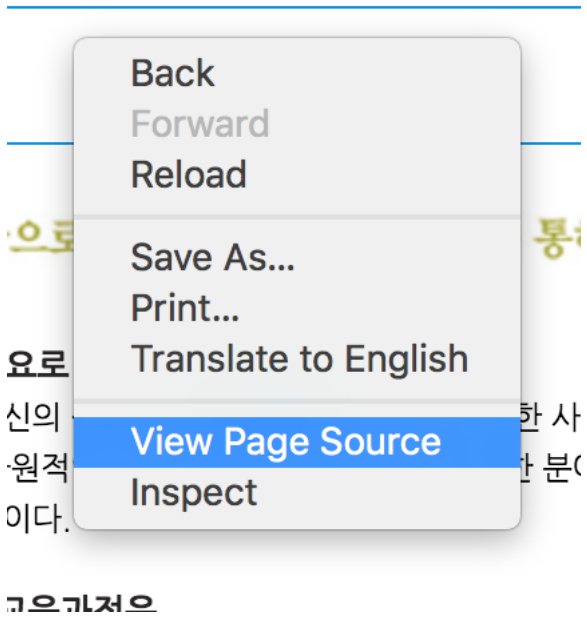
HTML, CSS, and Javascript



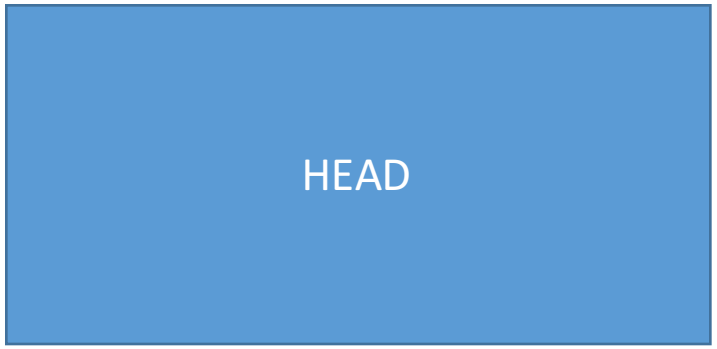
- HTML – House itself
- CSS – Painting the house and decoration
- Javascript – rich features, electricity on/off using a smartphone

How to Check HTML of a Web Page

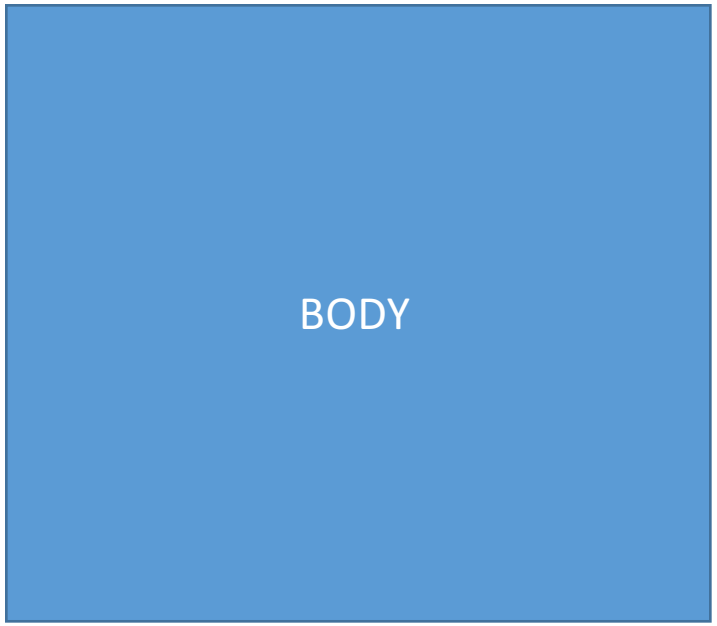
- In a Chrome web-browser, right click a page
- Select “View Page Source”



HTML Composition



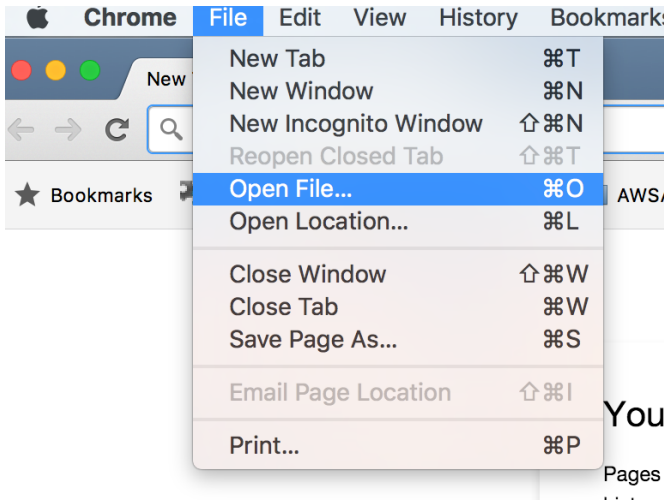
- Contains information about the page
- Title, meta (encoding), page styles, …



- The main contents shown in the page

HTML Getting Started

- Create your first webpage
- Open a text editor and type a text in the editor
  - “This is my first web page”
- Save it as a plain text with an extension of “html”
- Open Chrome web browser and load the html file



HTML Tags

- Tag/Element - Surround contents and apply meaning to them
  - Start tag with <> close tag with “/”
    - EX: <name>WebClient Computing</name>
- <!DOCTYPE html> : document type declaration
- <html>…</html> : The start and beginning of html document
- <body>...</body> : the main content in the browser window

```
<!DOCTYPE html>
<html>
<body>
  WebClient Computing
</body>
</html>
```

More about HTML DOCTYPE

- Specify which version of HTML was used to describe the content
- Helps a browser to correctly render a page
- HTML5
  - <!DOCTYPE html>
  - We will use this in the class
- HTML 4.01
  - <!DOCTYPE HTML PUBLIC "-//W3C//DTD HTML 4.01//EN" "http://www.w3.org/TR/html4/strict.dtd">

Different Capabilities across Different HTML

Tag	HTML5	Transitional	Strict	Frameset	XHTML 1.1
<a>	Yes	Yes	Yes	Yes	Yes
<abbr>	Yes	Yes	Yes	Yes	Yes
<acronym>	No	Yes	Yes	Yes	Yes
<address>	Yes	Yes	Yes	Yes	Yes
<applet>	No	Yes	No	Yes	No
<area>	Yes	Yes	Yes	Yes	No
<article>	Yes	No	No	No	No
<aside>	Yes	No	No	No	No
<audio>	Yes	No	No	No	No

http://www.w3schools.com/tags/ref_html_dtd.asp

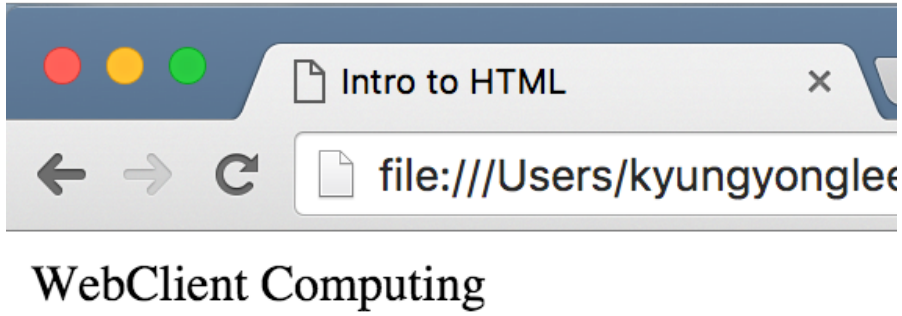
HTML Comments

- HTML comments
  - Start with <!--
  - End with -->

HTML Page Titles

- head and title tag
  - Contains information about the page (metadata – information about the data)
  - Title information appears in the top-left corner in a browser (Tab)
  - Title is also used in favorites or bookmarks page

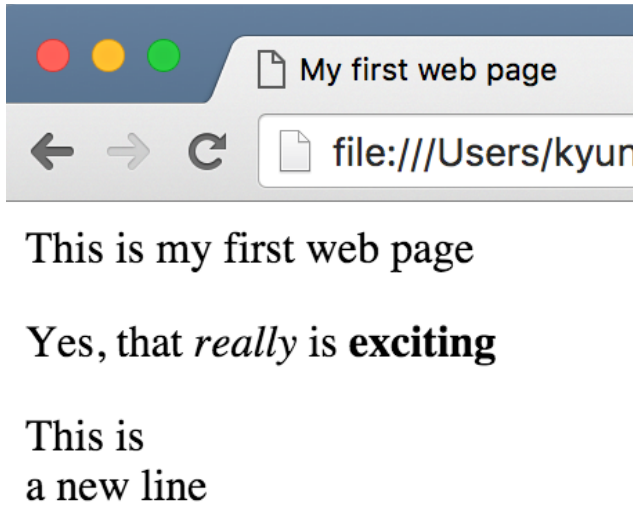
```
<!DOCTYPE html>
<html>
<head>
  <title>Intro to HTML</title>
<head>
<body>
  WebClient Computing
</body>
</html>
```



HTML Tags – p, em, strong

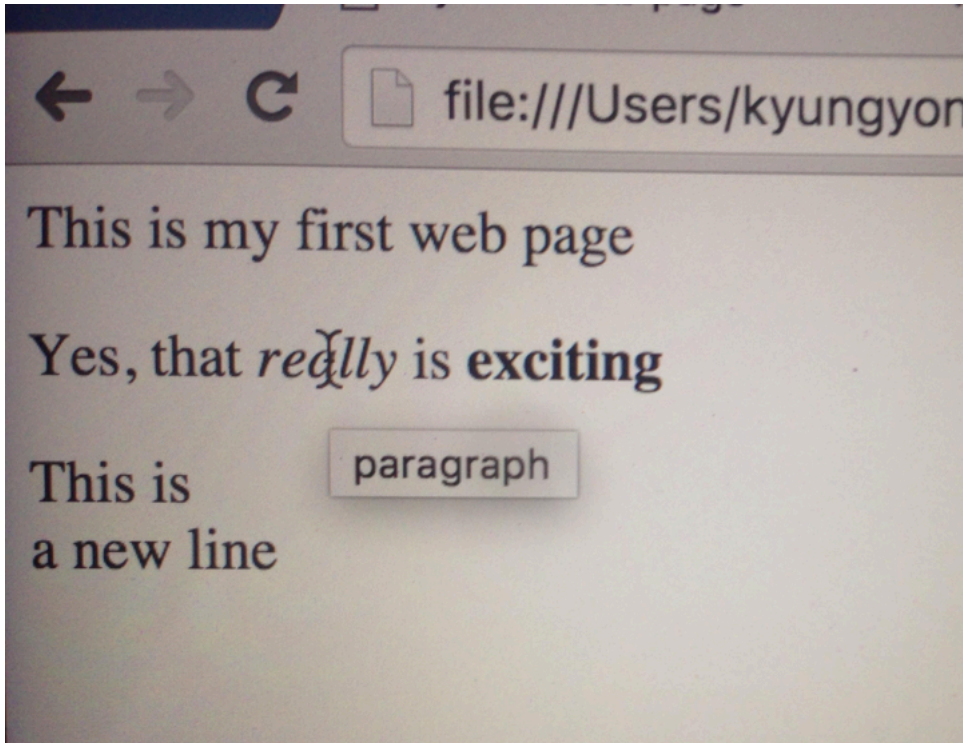
- p – separates ”pragraph”
- em – emphasize content using *italics*
- strong – make content **bold**
- br – insert a new line

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
</head>
<body>
  This is my first web page
  <p>Yes, that <em>really</em>
  is <strong>exciting</strong></p>
  This is <br> a new line
</body>
</html>
~
```



HTML Tag Attributes

- Element is an independent object
- Attribute
  - Provide additional information of the element
  - Should be located within the starting tag
  - Key-value format (name=“value”)
  - Value is recommended to be quoted
- Example
  - Element: Computer
  - Attribute: Laptop, desktop, workstation, …



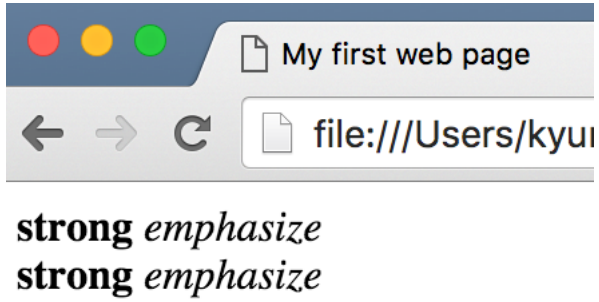

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
</head>
<body>
  This is my first web page
  <p title="paragraph">Yes, that <em>really</em>
  is <strong>exciting</strong></p>
  This is <br> a new line
</body>
</html>
```

HTML Semantics and Style

- implies the element is **strongly** important
- implies emphasize (강조)
- The effect of is same as <b> : **Bold**
- The effect of is same as <i> : *Italic*
- The difference
  - <b> and <i> is simply styling
  - and is implication/semantic (의미부여)
  - For blind people, <b> and <i> cannot take effect
  - In mobile device, <b> might not be recognizable

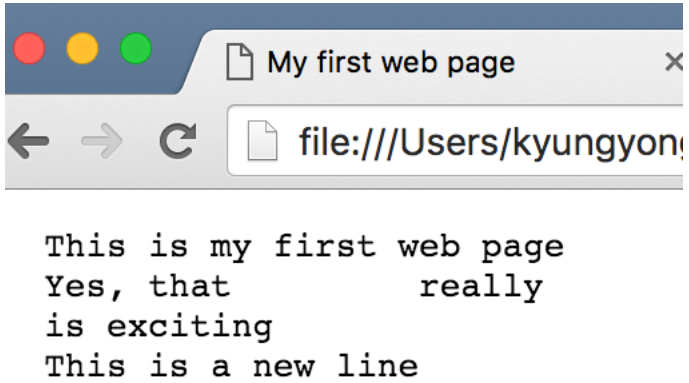
<b> <i>

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
</head>
<body>
  <strong>strong</strong> <em>emphasize</em><br>
  <b>strong</b> <i>emphasize</i>
</body>
</html>
```



HTML Display

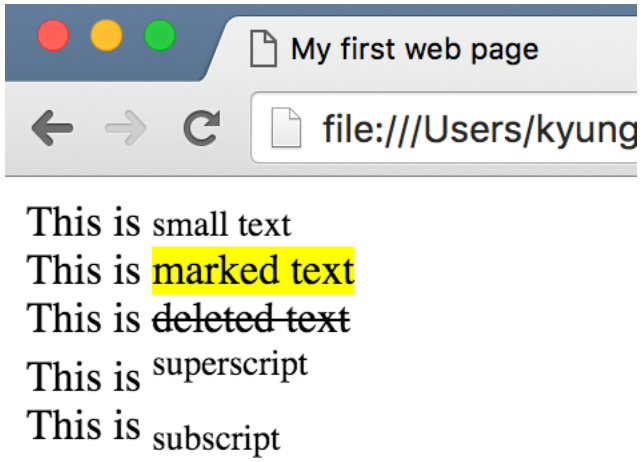
- HTML developer does not know how the html is rendered in different browsers
  - Adding extra spaces and lines do not work
  - Browser will remove it – it depends on the browser
  - Use explicit command - ,
- To show it as it is in the html, use <pre></pre>




```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
</head>
<body>
  <pre>
  This is my first web page
  Yes, that         really
  is exciting
  This is a new line
</body>
</html>
```

small, mark, deleted, superscript, subscript

- Small
  - Small text: This is <small>small text</small>
- Mark
  - Marked text: This is <mark>marked text</mark>
- Del
  - Deleted text: This is <del>deleted text</del>
- Sup
  - Superscript: This is ^{superscript}
- Sub
  - Subscript: This is _{subscript}

Examples

```
<!DOCTYPE html>
<html>
<head>
  <title>My first web page</title>
</head>
<body>
This is <small>small text</small><br>
This is <mark>marked text</mark><br>
This is <del>deleted text</del><br>
This is <sup>superscript</sup><br>
This is <sub>subscript</sub>

</body>
</html>
```